Research Summary

Piecewise Linear Representation Theory of Equivariant Neural Networks

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Motivation

My Master's thesis studies equivariant neural networks using representation theory. Suppose that a finite group G acts on the vector spaces appearing as layers of a neural network. After decomposing these spaces into irreducible G-representations, Schur's lemma determines which irreducible components can communicate through a linear equivariant layer. In particular, non-isomorphic irreducibles cannot interact linearly.

A neural network is not, however, a composition of linear maps alone. Between its linear layers, it applies nonlinear activation functions such as $\mathrm{ReLU}(x) = \max(x, 0)$. These activations are often piecewise linear and can create interactions between non-isomorphic irreducible representations that are forbidden at the linear level.

Earlier work of Gibson, Tubbenhauer and Williamson records these nonlinear interactions in an interaction graph Γ. Its vertices are the irreducible summands of the representation, and an edge $L \to K$ is present when the corresponding block of the activation is nonzero.

Their piecewise linear analogue of Schur's lemma gives the existence criterion

$$\mathrm{Hom}_G^{\mathrm{PL}}(L, K) \neq 0 \quad \Longleftrightarrow \quad \ker L \subseteq \ker K,$$

where $\mathrm{Hom}_G^{\mathrm{PL}}(L, K)$ denotes the space of G-equivariant piecewise linear maps from L to K.

For a fixed piecewise linear map F, this gives only the upper bound

$$L \to K \text{ is an edge of } \Gamma_F \quad \Longrightarrow \quad \ker L \subseteq \ker K.$$

Conversely, the kernel inclusion states that the edge is realised by *some* equivariant piecewise linear map, not necessarily by the particular activation under consideration. It does not determine

- the algebraic degree at which the interaction first becomes possible,
- whether a particular activation, such as ReLU or absolute value, actually realises it,
- which interactions can be removed from an architecture without affecting its symmetry-compatible expressive possibilities, or
- how nonlinear frequency mixing should be organised in a general equivariant architecture.

My thesis develops a graded and activation-sensitive refinement of this interaction graph.

From piecewise linear maps to symmetric powers

Polynomial equivariant maps are familiar representation theoretic objects. A homogeneous polynomial map $L \to K$ of degree k is equivalent to a linear equivariant map from the k-th symmetric power:

$$\mathrm{Poly}_k(L, K)^G \cong \mathrm{Hom}_G(\mathrm{Sym}^k L, K).$$

Thus, if an activation were polynomial, its interactions could be analysed degree by degree using ordinary character theory, symmetric powers and covariants. The difficulty is that piecewise linear activations are not polynomial. The main technical idea is therefore to associate with a piecewise linear map an infinite sequence of polynomial equivariant components.

Let L be a real G-representation equipped with a G-invariant inner product, and let γ_L be the standard Gaussian probability measure on L:

$$d\gamma_L(x) = (2\pi)^{-\dim(L)/2} e^{-\|x\|^2/2}\, dx.$$

The following properties are crucial:

- A piecewise linear map has at most linear growth and is therefore square-integrable with respect to γ_L.
- The measure γ_L is invariant under $O(L)$ and hence under the action of G.
- The orthogonal polynomials for γ_L are the Hermite polynomials, whose degree-k component carries the k-th symmetric-power representation.

Consequently, an equivariant piecewise linear block $F\colon L \to K$ may be regarded as an element of the Hilbert space

$$L^2(\gamma_L; K) = \left\{ F\colon L \to K : \int_L \|F(x)\|^2 \, d\gamma_L(x) < \infty \right\}.$$

It admits the orthogonal decomposition

$$L^2(\gamma_L; K) = \widehat{\bigoplus}_{k\geq 0} \mathcal{H}_k(L) \otimes K,$$

where $\mathcal{H}_k(L)$ is the k-th Wiener chaos, represented by degree-k Hermite polynomials. As an orthogonal-group representation, $\mathcal{H}_k(L) \cong \mathrm{Sym}^k(L^*)$ which after identifying $L \cong L^*$, gives

$$\big(\mathcal{H}_k(L) \otimes K\big)^G \cong \mathrm{Hom}_G(\mathrm{Sym}^k L, K).$$

Therefore, every equivariant piecewise linear block has an orthogonal expansion

$$F = \sum_{k\geq 0} \Pi_k F, \qquad \Pi_k F \in \mathrm{Hom}_G(\mathrm{Sym}^k L, K).$$

The Gaussian measure should therefore be viewed as a bridge from a non-polynomial equivariant map to a sequence of classical representation theoretic objects.

The degree refinement

For irreducible real G-representations L and K, define the degree set

$$D(L,K) := \left\{ k \geq 0 : \mathrm{Hom}_G(\mathrm{Sym}^k L, K) \neq 0 \right\},$$

and, whenever $D(L,K) \neq \varnothing$, define $d(L,K) := \min D(L,K)$. One of the main structural results of the thesis is

$$\ker L \subseteq \ker K \quad \Longleftrightarrow \quad D(L,K) \neq \varnothing \quad \Longleftrightarrow \quad \mathrm{Hom}_G(\mathrm{Sym}^k L, K) \neq 0 \text{ for some } k \geq 0.$$

Thus, the kernel-inclusion graph is obtained from the graded theory by forgetting the degree. The integer $d(L,K)$ refines each permitted edge by recording the first symmetric power in which the target representation appears. This relates the kernel poset of irreducible representations to the graded covariant spaces $\mathrm{Hom}_G(\mathrm{Sym}^k L, K)$. The problem of nonlinear interaction is hence converted into the study of symmetric powers, covariant modules, character multiplicities and their degree structure.

Activation classification

The abstract degree set determines which interactions are permitted by representation theory, but a particular coordinatewise activation need not realise every permitted degree. Suppose that $\sigma\colon \mathbb{R} \to \mathbb{R}$ acts coordinatewise in a G-stable permutation basis. Under the positive-homogeneity assumptions, the degree-k component of the block from L to K factors as

$$\Pi_k F_L^K = a_k(\sigma)\, M_k^{K,L},$$

where $a_k(\sigma) = \frac{1}{k!}\mathbb{E}\left[\sigma(U)\,\mathrm{He}_k(U)\right]$, with $U \sim \mathcal{N}(0,1)$, is the k-th Hermite coefficient of the scalar activation, while $M_k^{K,L} \in \mathrm{Hom}_G(\mathrm{Sym}^k L, K)$ depends only on the representation and the chosen basis.

This separates two distinct questions:

- $\mathrm{Hom}_G(\mathrm{Sym}^k L, K) \neq 0$ asks whether degree k is abstractly permitted,
- $a_k(\sigma) M_k^{K,L} \neq 0$ asks whether the chosen activation and basis actually realise that interaction.

The interaction graph of the activation is therefore determined by the intersection of its Hermite spectrum $\mathrm{supp}\, a_\bullet(\sigma) = \{k \geq 0 : a_k(\sigma) \neq 0\}$ with the degrees realised by the universal maps $M_k^{K,L}$.

Every positively homogeneous piecewise linear scalar activation has the form

$$\sigma(x) = \frac{m_+ + m_-}{2} x + \frac{m_+ - m_-}{2} |x|, \qquad m_\pm \in \mathbb{R}.$$

The relevant Hermite supports are

$$\mathrm{supp}\, a_\bullet(\mathrm{id}) = \{1\}, \quad \mathrm{supp}\, a_\bullet(|\cdot|) = \{0,2,4,6,\ldots\}, \quad \mathrm{supp}\, a_\bullet(\mathrm{ReLU}) = \{0,1,2,4,6,\ldots\}.$$

It follows that positively homogeneous piecewise linear activations produce at most three interaction graphs:

- the linear graph, consisting of the degree-one interactions,
- the absolute-value graph, consisting of the even-degree interactions,
- the ReLU-type graph, consisting of the even-degree interactions together with degree one.

Consequently, ReLU and absolute value have the same off-diagonal interaction graph, and their only possible difference lies on the diagonal. This is significant because ReLU and absolute value are analytically different activations, but from the perspective of interactions between distinct irreducible components, they generate the same structure. The off-diagonal flow of information is therefore determined by the even-degree part of their Hermite spectra, while the linear component of ReLU contributes only the identity interactions. More broadly, the factorisation

$$\Pi_k F_L^K = a_k(\sigma; s) M_k^{K,L}$$

provides a practical way to compare activation functions without recomputing the representation theory of the architecture. The universal maps $M_k^{K,L}$ can be computed once, while changing the activation changes only the scalar coefficients $a_k(\sigma; s)$. This makes it possible to determine which parts of an equivariant architecture are genuinely affected by the choice of activation, to identify interactions that are redundant across different activations, and potentially to design or simplify architectures using algebraic information before training.

For finite abelian groups, we lay out a closed-form description of the degree set in terms of the character group, which also confirms results from the previous paper, while now providing a graded non-Boolean picture.

Obtaining comparable closed forms for non-abelian groups is substantially more difficult. Their irreducible representations are generally higher-dimensional, and symmetric powers no longer reduce to products of scalar characters. One must instead determine the multiplicities $\dim \mathrm{Hom}_G(\mathrm{Sym}^k L, K)$, which are plethysm problems and can depend on detailed character and tensor-product data. Multiplicities may also occur, and the occurrence of K in $\mathrm{Sym}^k L$ need not follow a simple congruence rule. For this reason, the abelian case admits uniform arithmetic formulae, whereas non-abelian families are more likely to require group-specific invariant theory, character computations or new structural methods.

Open problems and future directions

- **Compare abstract and coordinatewise realisability.** Determine when

  $$\mathrm{Hom}_G(\mathrm{Sym}^k L, K) \neq 0$$

  guarantees that the coordinatewise universal map $M_k^{K,L}$ is nonzero. This is particularly important for permutation representations on non-free orbits.

- **Treat multiplicities.** Extend the block-level theory to representations in which the same irreducible occurs with multiplicity, where individual interaction graphs depend on choices of inclusions and projections.

- **Compute first admissible degrees.** Develop general formulae or effective bounds for $d(L, K)$ using covariant modules, invariant theory, character theory and the geometry of finite-group orbits.

- **Extend beyond finite groups.** Formulate analogous interaction and degree invariants for compact groups and homogeneous spaces.

- **Establish learning-theoretic consequences.** Determine when degree-restricted architectures provide provable improvements in parameter efficiency, approximation complexity, generalisation or robustness.